

HORNSBY GIRLS' HIGH SCHOOL



2007 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 1

General Instructions

- Reading Time- 5 minutes
- Working Time – 2 hours
- Write using a black or blue pen
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown for every question.
- Begin each question on a fresh sheet of paper.

Total marks (84)

- Attempt Questions 1-7
- All questions are of equal value

Total Marks – 84

Attempt Questions 1-7

All Questions are of equal value

Begin each question on a NEW SHEET of paper, writing your name and question number at the top of the page. Extra paper is available.

Question 1 (12 marks) Use a SEPARATE sheet of paper. **Marks**

- | | |
|--|---|
| (a) Let A (3,4) and B (-2,5) be points in the plane. Find the co-ordinates of the Point C which divides the interval externally in the ratio 1:3. | 1 |
| (b) A committee of 4 boys and 4 girls is chosen from a class of 8 boys and 6 girls | |
| (i) Calculate the probability that the committee includes the eldest boy and excludes the eldest girl | 2 |
| (ii) Find the number of ways the committee of 4 boys and 4 girls can be arranged in a circle so that each of the girls are separated. | 1 |
| (c) For the function $f(x) = \frac{1}{2} \cos^{-1} \frac{1}{3}x$ | |
| (i) State the range and the domain of $f(x)$ | 2 |
| (ii) Sketch the graph of $y = f(x)$ | 1 |
| (d) Using the expansion of $\sin(A-B)$ or otherwise prove the exact value of $\sin \frac{\pi}{12} = \frac{\sqrt{6}-\sqrt{2}}{4}$ | |
| (e) The curves $y = \log_e x$ and $y = -x^2 + 1$ intersect at the point P (1,0). Find the acute angle between the tangents to the curves at the point P. Give your answer to the nearest degree. | 3 |

Question 2 (12 marks) Use a SEPARATE sheet of paper.

Marks

- (a) Solve $|x-1| \leq |x+1|$ **2**
- (b) Express $\cos\left(2 \sin^{-1} \frac{a}{b}\right)$ in terms of a and b . **2**
- (c) The graph of $y = \sin x$ for $\frac{\pi}{12} \leq x \leq \pi$ is rotated about the x axis. Calculate the volume of the solid generated. **3**
- (d) Write down the general solution of the equation $\sqrt{3} \cos 2x - \sin 2x = 2$ **3**
- (e) In how many ways can the letters of the word GEOLOGIST be arranged so that the letters G will be together? **2**

Question 3 (12 marks) Use a SEPARATE sheet of paper.

Marks

- a) Calculate $\int_0^{\sqrt{\frac{27}{2}}} \frac{1}{9+2x^2} dx$ **3**
- (b) Calculate the area between the curve $y = \sin^{-1} x$, the x axis and the line $x=1$ **2**
- (c) Prove $\frac{\sin 2\theta + \sin \theta}{1 + \cos 2\theta + \cos \theta} = \tan \theta$ **2**
- (d) An eight sided die has 5 green faces and 3 blue faces. If the die is tossed 100 times find the most likely number of green faces and the probability of this occurring (correct to 3 decimal places) **3**
- (e) Find the value of n , if the coefficients of x^5 and x^6 in the expansion of $(3 + 2x)^n$ have the same value **2**

Question 4 (12 marks) Use a SEPARATE sheet of paper. **Marks**

- (a) A particle travelling in a straight line is governed by the equation $v^2 = 15 + 2x - x^2$ where v is the velocity in m/s and x is the distance travelled in time t seconds.
- (i) Prove that the particle undergoes simple harmonic motion **1**
- (ii) (1) Find the centre of the motion **1**
 (2) Find the amplitude of the motion **1**
 (3) Find the period of the motion **1**
- (iii) Write down the maximum speed and the maximum acceleration **2**
- (iv) Given that the particle was originally at its equilibrium position write down an equation for the position $x = f(t)$ and hence or otherwise find the velocity when $t = \frac{\pi}{4}s$ **3**
- (b) Prove by Mathematical Induction that:
 $2(1!) + 5(2!) + 10(3!) + \dots + (n^2 + 1)n! = n(n+1)!$
 for all positive integers $n \geq 1$ **3**

Question 5 (12 marks) Use a SEPARATE sheet of paper. **Marks**

- (a) The Polynomial $2x^3 + ax^2 + bx + 6$ has $(x - 1)$ as a factor and leaves a remainder of -12 when divided by $(x - 2)$. Find the values of a and b . **2**
- (b) Solve the equation $x^3 + 2x^2 - 5x - 6 = 0$ given that one of its roots is equal to the sum of the other two roots **2**
- (c) Two straight roads intersect at right angles. At a given instant a car is 30 km from the intersection and is travelling towards it at 50 km/h while the truck is 40 km from the intersection and is travelling away from it at 40 km/h.. At what rate is the direct distance between them changing at this instant? **2**
- (d) Show that $\frac{d}{dx} [\sin^{-1}(\frac{1}{2}\sin x)] = \frac{\cos x}{\sqrt{4 - \sin^2 x}}$
 hence evaluate $\int_0^{\frac{\pi}{2}} \frac{\cos x}{\sqrt{4 - \sin^2 x}} dx$ **3**
- (e) Using the substitute $u = 1 - 3x$
 evaluate $\int_0^{\frac{1}{3}} 3x(1 - 3x)^4 dx$ **3**

Question 6 (12 marks) Use a SEPARATE sheet of paper.

Marks

(a) A rocket is fired at a speed of V m/s at an angle θ to the horizontal where $\tan \theta = \frac{4}{3}$. Neglecting air resistance and using acceleration due to gravity as $g = 10\text{m/s}^2$

(i) Show that the horizontal position x and the vertical position y of the rocket at any time t is given by

$$x = \frac{3Vt}{5} \text{ and } y = \frac{4Vt}{5} - 5t^2$$

2

(ii) The rocket hits a target which has the coordinates (324,27). Find the value of V

3

(b) Using the expansion of $(1+x)^n$

$$\text{Show } \frac{-1}{n+1} = -{}^nC_0 + \frac{1}{2}({}^nC_1) - \frac{1}{3}({}^nC_2) + \dots + \frac{(-1)^{n+1}}{n+1}({}^nC_n)$$

3

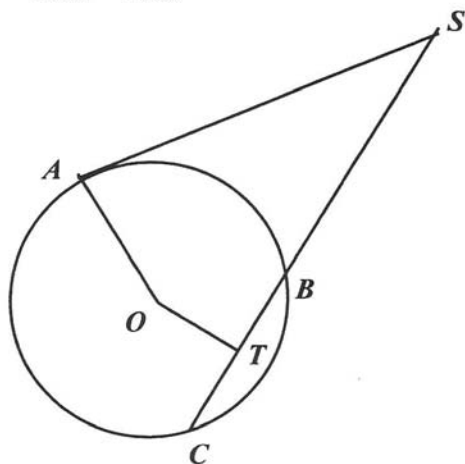
(c) A, B, C are three points on the circumference of a circle, centre O . The tangent at A meets CB produced at S . T is the mid point of BC .

Prove that

- (i) $TOAS$ is a cyclic quadrilateral
(ii) $\angle OAT = \angle OST$

2

2



Question 7 (12 marks) Use a SEPARATE sheet of paper.

Marks

(a) The tangent at $P(2ap, ap^2)$ To the parabola $x^2 = 4ay$ meets the x axis at A .

(i) Find the co-ordinates of A

2

(ii) If S is the Focus of this parabola. Prove SA is perpendicular to AP

1

(iii) Show that the equation of the locus of the Centre C of the circle which passes through the three points P, S and A is a parabola and write down the coordinates of its vertex

3

(b) The rate at which a body warms or cools in air is proportional to the difference between its temperature T and the constant temperature of its surroundings S . The temperature obeys the differential equation $\frac{dT}{dt} = k(T - S)$. You may assume the solution $T = S + Ae^{kt}$

(i) A cup of boiling water at 100°C and a cup of iced water at 0°C are placed simultaneously in a room which has a temperature of 25°C . After 5 minutes the temperature of the boiling water has fallen to 55°C and the temperature of the iced water has risen to 15°C . Find the time at which the temperature of the two liquids differs by 10°C .
(Give your answer correct to two decimal places)

4

(ii) Draw a graph of the behaviour of the temperature T of both liquids as t becomes large.

2

End of Examination

Question 1 (12 marks)

1) $A(3,4)$ and $B(-2,5)$

$$x = \frac{2+9}{2} \quad y = \frac{5+1}{2}$$

$$x = 5\frac{1}{2} \quad y = 3\frac{1}{2}$$

$$C(5\frac{1}{2}, 3\frac{1}{2})$$

8 boys 6 girls

4 boys 4 girls

1) $n(s) = 8C_4 \times 6C_4$

$$= 70 \times 15$$

$$= 1050$$

$n(e) = 7C_4 \times 5C_4$

$$= 35 \times 5$$

$$= 175$$

$P(e) = \frac{175}{1050}$

$$= \frac{1}{6}$$

$$3! \times 4!$$

$$= 144$$

$$f(x) = \frac{1}{2} \cos^{-1} \frac{1}{2}x$$

$$2y = \cos^{-1} \frac{x}{2}$$

Range

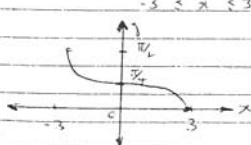
$$0 < 2y \leq \pi$$

$$0 \leq y \leq \frac{\pi}{2}$$

Domain

$$-1 \leq \frac{x}{2} \leq 1$$

$$-2 \leq x \leq 2$$



(d)

$$\sin(A-B) = \sin A \cos B - \cos A \sin B$$

$$\sin(\frac{\pi}{3} - \frac{\pi}{4}) = \sin \frac{\pi}{3} \cos \frac{\pi}{4} - \cos \frac{\pi}{3} \sin \frac{\pi}{4}$$

$$\sin \frac{\pi}{12} = \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} - \frac{1}{2} \times \frac{1}{\sqrt{2}}$$

$$= \frac{\sqrt{3}-1}{2\sqrt{2}}$$

$$= \frac{\sqrt{3}-1}{2\sqrt{2}}$$

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$$= \frac{\sqrt{3}-1}{2\sqrt{2}}$$

$$= \frac{\sqrt{3}-1}{2\sqrt{2}}$$

(e) $f(x) = \ln x$

$$f(x) = \frac{1}{2}$$

$$f(1) = 1$$

$$m_1 = 1$$

$$f(x) = -x^2 + 1$$

$$f(x) = -2x$$

$$f(1) = -2$$

$$m_2 = -2$$

$$\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$$

For acute angle θ

$$\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$$

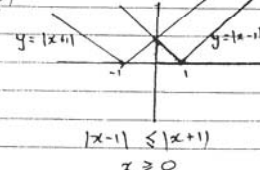
$$\tan \theta = \frac{1+2}{1-2}$$

$$\tan \theta = 3$$

$$\theta = 72^\circ \text{ (Nearest degree)}$$

Question 2 12 marks

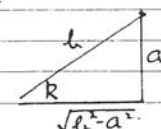
(a)



1) $\cos(2 \sin^{-1} \frac{a}{c})$

Let $\sin^{-1} \frac{a}{c} = k$

$$\sin k = \frac{a}{c}$$



$$\cos 2k = 2 \cos^2 k - 1$$

$$= 2(\frac{b^2-a^2}{c^2}) - 1$$

$$= \frac{2b^2-2a^2-b^2}{c^2}$$

$$= \frac{b^2-2a^2}{c^2}$$

$$= \frac{b^2-2a^2}{c^2}$$

$$V = \pi \int_0^{\pi} y^2 dx$$

$$= \pi \int_0^{\pi} \sin^2 x dx$$

$$= \frac{\pi}{2} \int_0^{\pi} 1 - \cos 2x dx$$

$$= \frac{\pi}{2} [x - \frac{1}{2} \sin 2x]_0^{\pi}$$

$$= \frac{\pi}{2} [\pi - 0] - [\frac{\pi}{2} - \frac{1}{4}]$$

$$= \frac{\pi}{2} [\frac{11\pi}{12} + \frac{1}{4}]$$

$$= \frac{\pi}{2} [\frac{11\pi+3}{12}]$$

$$= \frac{\pi}{24} [11\pi+3] \text{ cubic units}$$

(d)

$$\sqrt{3} \cos 2x - 1 \sin 2x = 2$$

use Subsidiary angle

$$R = \sqrt{4} \quad \tan \theta = \frac{1}{\sqrt{3}}$$

$$R = 2 \quad \theta = \frac{\pi}{6}$$

$$\therefore 2 \cos(2x + \frac{\pi}{6}) = 2$$

$$\cos(2x + \frac{\pi}{6}) = 1$$

$$\cos(2x + \frac{\pi}{6}) = \cos 0$$

$$\therefore 2x + \frac{\pi}{6} = 2n\pi \pm 0$$

$$2x = 2n\pi - \frac{\pi}{6}$$

$$x = n\pi - \frac{\pi}{12}$$

(e) $EEO LIST$

$$EEO LIST$$

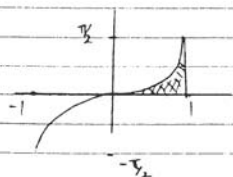
$$EEO LIST$$

$$= \frac{8!}{2!}$$

$$= 20160$$

Question 3

$$\begin{aligned} \text{(a)} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{9+2x^2} dx &= \frac{1}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{\frac{9}{2}+x^2} dx \\ &= \frac{1}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{(\frac{3}{\sqrt{2}})^2 + x^2} dx \\ &= \frac{1}{2} \times \frac{\sqrt{2}}{3} \left[\tan^{-1} \frac{\sqrt{2}x}{3} \right]_0^{\frac{\pi}{2}} \\ &= \frac{\sqrt{2}}{6} \left[\tan^{-1} \frac{\sqrt{2} \times \frac{\pi}{2}}{3} - \tan^{-1} 0 \right] \\ &= \frac{\sqrt{2}}{6} \left[\tan^{-1} \sqrt{3} \right] \\ &= \frac{\sqrt{2}}{6} \times \frac{\pi}{3} \\ &= \frac{\sqrt{2} \pi}{18} \end{aligned}$$



$$\begin{aligned} A &= \frac{\pi}{2} - \int_0^{\frac{\pi}{2}} \sin x dx \\ &= \frac{\pi}{2} - \left[-\cos x \right]_0^{\frac{\pi}{2}} \end{aligned}$$

$$\begin{aligned} &= \frac{\pi}{2} - (0+1) \\ &= \frac{\pi}{2} - 1 \text{ sq units.} \end{aligned}$$

$$\begin{aligned} \text{(c)} \text{ LHS} &= 2 \sin \theta \cos \theta + \sin \theta \\ &= 1 + 2 \cos^2 \theta - 1 + \cos \theta \\ &= \frac{\sin \theta (2 \cos \theta + 1)}{\cos \theta (2 \cos \theta + 1)} \\ &= \tan \theta \\ &= \text{RHS} \end{aligned}$$

$$\begin{aligned} \text{(d)} \text{ Let } g &= \frac{x}{y} \\ h &= \frac{y}{x} \\ \text{max Coef} &= \frac{n-r+1}{r} \frac{\frac{x}{y}}{\frac{y}{x}} \geq 1 \\ \frac{101-r}{r} \frac{3}{5} &\geq 1 \\ 303-3r &\geq 5r \\ 8r &\leq 303 \\ r &\leq 37.875 \\ r &= 37 \end{aligned}$$

$$\begin{aligned} \therefore \text{MAX Coefficient} &= \frac{63}{37} \left(\frac{3}{5} \right) \\ \text{(e) MOST likely green faces is 63} \\ \text{Probability of 63 green faces} &= \left(\frac{100}{37} \right)^{63} \left(\frac{3}{5} \right)^{37} \\ &= 0.082 \end{aligned}$$

Term in x^5

$$n_c \frac{n-5}{3} \frac{5}{2}$$

Term in x^6

$$n_c \frac{n-6}{3} \frac{6}{2}$$

$$n_c \frac{n-5}{3} \frac{5}{2} = n_c \frac{n-6}{3} \frac{6}{2}$$

$$n_c \div n_c = \frac{3}{3}$$

$$\frac{1}{5!} \times \frac{(n-6)! 6!}{n!} = \frac{3}{3}$$

$$\frac{6}{5!} = \frac{3}{3}$$

$$2n-10=18$$

$$\begin{aligned} 2n &= 28 \\ n &= 14 \end{aligned}$$

Question 4

$$v^2 = 15 + 2x - x^2$$

$$x = \frac{d}{dt} \left(\frac{1}{2} x^2 + x - \frac{1}{3} x^3 \right)$$

$$= 1 - x$$

$$= -1(x-1)$$

$$= -n^*(x-h) \text{ where } n=1, h=1$$

Simple harmonic motion
Centre is at $x=1$

$$\text{(a) Centre is at } x=1$$

$$\text{(p) Particle stops } x^2 - 2x - 15 = 0$$

$$(x-5)(x+3) = 0$$

$$x=5 \text{ and } x=-3$$

$$\therefore \text{Amplitude} = 4 \text{ units.}$$

$$\begin{aligned} \text{(x) Period } T &= \frac{2\pi}{\omega} \\ &= \frac{2\pi}{2\pi} \text{ seconds} \end{aligned}$$

(iii) MAX velocity is at $x=1$

$$\text{(e) } v^2 = 15 + 2 - 1$$

$$v^2 = 16$$

$$v = \pm 4$$

$$\text{MAX Speed is } 4 \text{ m/s.}$$

$$\text{MAX Acc is at } x=5 \text{ or } x=-3$$

$$\text{acc} = -1(5-1) \text{ or } -1(-3-1)$$

$$\text{MAX Acc} = \pm 4 \text{ m/s}^2$$

$$\text{When } t = \pi$$

$$\begin{aligned} \text{vel} &= 4 \times \frac{1}{2} \\ &= 2\sqrt{2} \text{ m/s} \end{aligned}$$

(b) When $n=1$

$$\text{LHS} = 2 \times 1!$$

$$= 2$$

$$\text{RHS} = 1(2)!$$

$$= 2$$

$$\therefore \text{TRUE FOR } n=1$$

Assume TRUE FOR $n=k$

$$\text{(a) Assume}$$

$$2(1!) + 5x(2!) + \dots + (k+1)k! = (k+1)!$$

$$\text{RHS}$$

$$2(1!) + 5x(2!) + \dots + (k+1)k! = (k+1)!$$

$$\text{LHS} = \frac{1}{2}(k+1)! + \frac{1}{2}(k+1)!(k+1)$$

$$= \frac{1}{2}(k+1)! + \frac{1}{2}(k+1)!(k+1)$$

$$= \frac{1}{2}(k+1)! [1 + (k+1)]$$

$$\begin{aligned}
 &= (k+1) k! \{k + k^2 + k + 2\} \\
 &= (k+1) k! \{k^2 + 3k + 2\} \\
 &= (k+1) k! (k+1)(k+2) \\
 &= (k+1) (k+2)! \\
 &= \text{RHS}
 \end{aligned}$$

True for $n = k+1$

Since True for $n=1$ Then

True for $n=2, 3, 4$ all $n \geq 1$

Question 5

a) Let $P(x) = 2x^3 + ax^2 + bx + 6$

$P(1) = 0$

(e) $2 + a + b + 6 = 0$

$a + b + 8 = 0 \quad \text{--- (1)}$

$P(2) = -12$

(e) $16 + 4a + 2b + 6 = -12$

$4a + 2b + 34 = 0$

$2a + b + 17 = 0 \quad \text{--- (2)}$

(3) - (1) $a + 9 = 0$

$a = -9$

$b = 1$

$x^3 + 2x^2 - 5x - 6 = 0$

Let roots be $\alpha, \beta, (\alpha + \beta)$

Sum: $2\alpha + 2\beta = -2$

$\alpha + \beta = -1 \quad \text{--- (1)}$

Product: $\alpha\beta(\alpha + \beta) = 6 \quad \text{--- (2)}$

From (1)

$\beta = -1 - \alpha$

into (2) $\alpha(-1-\alpha)(-1) = 6$

$\alpha + \alpha^2 = 6$

$\alpha^2 + \alpha - 6 = 0$

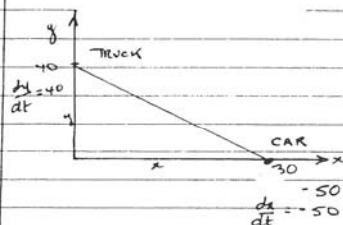
$(\alpha + 3)(\alpha - 2) = 0$

$\alpha = 3 \quad \alpha = 2 \quad \text{and } \alpha + \beta = -1$

$\beta = -3 \quad \beta = -2$

Roots $-3, 2, -1$

(c)



$\vec{r} = x\hat{i} + y\hat{j}$

$\frac{d\vec{r}}{dt} = x\frac{dx}{dt}\hat{i} + y\frac{dy}{dt}\hat{j}$

$\frac{dy}{dt} = \frac{30 \times -50}{50} + \frac{40 \times 40}{50}$

$= \frac{-1500 + 1600}{50}$

$= 2 \text{ km/h}$

(d)

$\frac{d}{dx} \sin^{-1}\left(\frac{1}{2}\sin x\right)$

$= \frac{1}{\sqrt{1 - \frac{1}{4}\sin^2 x}} \times \frac{1}{2} \cos x$

$= \frac{\cos x}{\sqrt{4 - \sin^2 x}}$

$\therefore \int_0^{\pi/2} \frac{\cos x \, dx}{\sqrt{4 - \sin^2 x}} = \left[\sin^{-1}\left(\frac{1}{2}\sin x\right) \right]_0^{\pi/2}$

$= \left[\sin^{-1}\left(\frac{1}{2}\right) - \left[\sin^{-1} 0 \right] \right]$

$= \frac{\pi}{6}$

$I = \int_0^3 3x(1-3x)^4 dx$

Let $u = 1-3x$

$du = -3dx$

$dx = -\frac{1}{3} du$

Initial Values

$x = \frac{1}{3} \Rightarrow u = 0$

$x = 0 \Rightarrow u = 1$

$I = -\frac{1}{3} \int_1^0 (1-u)u^4 du$

$= \frac{1}{3} \int_0^1 u^4 - u^5 du$

$= \frac{1}{3} \left[\frac{u^5}{5} - \frac{u^6}{6} \right]_0^1$

$= \frac{1}{3} \left[\frac{1}{5} - \frac{1}{6} \right] - [0]$

$= \frac{1}{90}$

Trajectory

$y = x \tan \theta - \frac{gx^2}{2v^2} (1 + \tan^2 \theta)$

where $\tan \theta = \frac{4}{3}$

$x = 324$

$y = 27$

$t = 5 \text{ s}$

$g = 10$

$27 = 324 \times \frac{4}{3} - 5 \times \frac{(324)^2}{v^2} \left(\frac{25}{9} \right)$

$9 \times 27 = 12 \times 324 - 125 \times \frac{324^2}{v^2}$

$243 = 3888 - 125 \times \frac{324^2}{v^2}$

$3645 v^2 = 125 \times 324^2$

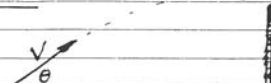
$v^2 = \frac{125 \times 324^2}{3645}$

$v = 60 \checkmark$

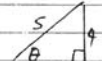
Question 6

(324, 27)

(b)



$\tan \theta = \frac{4}{3}$



Vertical

$\ddot{y} = -10$

$\dot{y} = -10t + \frac{4v}{5}$

$y = -5t^2 + \frac{4vt}{5}$

Horizontal

$\ddot{x} = 0$

$\dot{x} = \frac{3v}{5}$

$x = \frac{3vt}{5}$

$(1+x)^n = n_0 + n_1 x + n_2 x^2 + \dots + n_n x^n$

Integrate Both sides with respect to x

$\frac{(1+x)^{n+1}}{n+1} = n_0 x + \frac{1}{2} n_1 x^2 + \frac{1}{3} n_2 x^3 + \dots + \frac{n_n x^{n+1}}{(n+1)}$

Let $x=0$ to find k

$\frac{1}{n+1} = k$

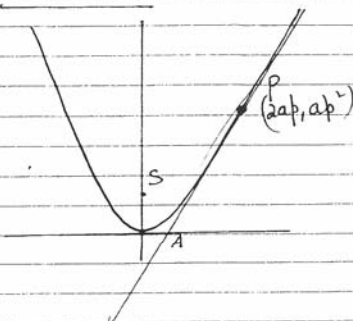
$\therefore \frac{(1+x)^{n+1}}{n+1} = n_0 x + \frac{1}{2} n_1 x^2 + \frac{1}{3} n_2 x^3 + \dots + \frac{n_n x^{n+1}}{(n+1)}$

Let $x = -1$

$\frac{-1}{n+1} = -n_0 + \frac{1}{2} n_1 - \frac{1}{3} n_2 + \dots + \frac{(-1)^{n+1} n_n}{(n+1)}$

3

Question 7



(i) Tangent at $(2ap, ap^2)$
 $y = px - ap^2$

Sub $y=0$

$px = ap^2$
 $x = ap$

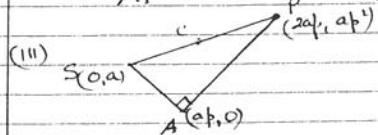
$\therefore A$ is $(ap, 0)$ (2)

(ii) $S(0, a)$ $A(ap, 0)$

Gradient $SA = \frac{a}{-ap}$
 $= -\frac{1}{p}$

Gradient $AP = p$

$\therefore AP \perp SA$ (1)



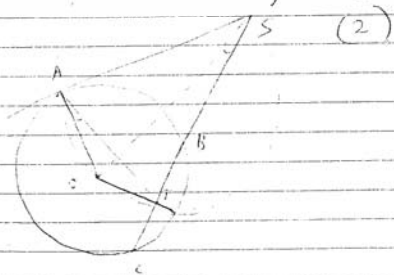
Centre of Circle $(ap, \frac{ap^2+a}{2})$

$x = ap$ $y = \frac{ap^2+a}{2}$

(i) $\angle SAO = 90^\circ$ (Tangent $\perp$ Radius)
 $\angle OTB = 90^\circ$ ($\because$ Line Joining Centre of Circle to meet point of contact is 90° Right)

$\therefore TOAS$ is a cyclic quad because opposite angles are supplementary (2)

$\angle OAT = \angle OST$ (angles at the circumference, standing on same arc OT) ($TOAS$ cyclic quad)



$p = \frac{x}{a}$

$y = \frac{ax^2}{a^2} + a$

$2ay = x^2 + a^2$

$x^2 = 2ay - a^2$

$x^2 = 2a(y-a)$ Locus

is the form of the general equation of a parabola

$(x-h)^2 = 4a(y-h)$

The Locus is the parabola

$(x-0)^2 = 2a(y-a)$

which has vertex $(0, a)$

(3)

(i) $T = S + Ae^{kt}$

$\frac{dT}{dt} = kAe^{kt}$

$= k(T-S)$

you may assume this in this question

(ii) BOILING WATER

$T = 25 + Ae^{kt}$

1] $100 = 25 + A$

2] $A = 75$

$\therefore T = 25 + 75e^{kt}$

5] $55 = 25 + 75e^{sk_1}$

$30 = 75e^{sk_1}$

$e^{sk_1} = \frac{2}{3}$

$k_1 = \frac{1}{s} \ln \frac{2}{3}$

$k_1 = \frac{1}{s} \ln \frac{2}{3}$

$T = 25 + 75e^{kt}$

iced water $k_2 t$

$T = 25 + Ae^{k_2 t}$

$T=0$ $0 = 25 + A$

$T=0$ $A = -25$

$T = 25 - 25e^{k_2 t}$

$T=15$ $15 = 25 - 25e^{sk_2}$

$t=5$ $-10 = -25e^{sk_2}$

$e^{sk_2} = \frac{2}{5}$

$k_2 = \frac{1}{s} \ln \frac{2}{5}$

$k_2 = \frac{1}{s} \ln \frac{2}{5}$

$T = 25 - 25e^{k_2 t}$

Difference in Temps = $10^\circ C$

$\frac{1}{s} \ln \frac{2}{5}$

$10 = (25 + 15e^{k_2 t}) - (25 - 25e^{k_2 t})$

$\frac{1}{s} \ln \frac{2}{5}$

$10 = 100e^{k_2 t}$

$\frac{1}{s} \ln \frac{2}{5}$

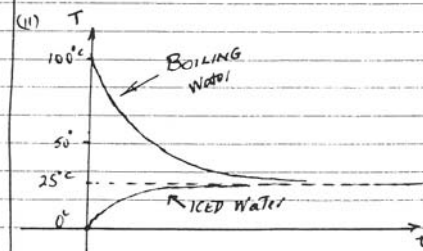
$1 = e^{k_2 t}$

$\frac{1}{s} \ln \frac{2}{5} = \ln 1$

$t = 5 \times \frac{\ln 1}{\ln \frac{2}{5}}$

$t = 12.56 \text{ mins}$

(4)



(2)